

Assessment tool 1 co-1

hackerank.com/challenges/java-stdin-and-stdout-1/problem?isFullScreen=true

HackerRank | Prepare | Java | Introduction | Java Stdin and Stdout I

Problem

Most HackerRank challenges require you to read input from `stdin` (standard input) and write output to `stdout` (standard output).

One popular way to read input from `stdin` is by using the `Scanner` class and specifying the Input Stream as `System.in`. For example:

```
Scanner scanner = new Scanner(System.in);
String myString = scanner.next();
int myInt = scanner.nextInt();
scanner.close();

System.out.println("myString is: " + myString);
System.out.println("myInt is: " + myInt);
```

The code above creates a `Scanner` object named `scanner` and uses it to read a `String` and an `int`. It then closes the `Scanner` object because there is no more input to read, and prints to `stdout` using `System.out.println(String)`. So, if our input is:

```
Hi 5
```

Our code will print:

```
myString is: Hi
myInt is: 5
```

Alternatively, you can use the `BufferedReader` class.

Submissions

Leaderboard

Discussions

Task

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Search

ENG IN

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Change Theme Language Java 15

```
1 import java.util.Scanner;
2
3 public class Solution {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         int a = scanner.nextInt();
8         int b = scanner.nextInt();
9         int c = scanner.nextInt();
10
11         System.out.println(a);
12         System.out.println(b);
13         System.out.println(c);
14
15         scanner.close();
16     }
17 }
18
19
20
```

Line: 20 Col: 1

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Run Code Submit Code

2.

hackerank.com/challenges/java-if-else/problem?isFullScreen=true

HackerRank | Prepare | Java | Introduction | Java If-Else

Problem

In this challenge, we test your knowledge of using `if-else` conditional statements to automate decision-making processes. An `if-else` statement has the following logical flow:

IF (A = TRUE)
Then B
Else C
End IF

```
graph TD
    Start(( )) --> A{A}
    A -- TRUE --> B([B])
    A -- FALSE --> C([C])
    B --> End((( )))
    C --> End
```

Submissions

Leaderboard

Discussions

Source: Wikipedia

Change Theme Language Java 15

```
1 import java.util.Scanner;
2
3 public class Solution {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int N = sc.nextInt();
7         if (N % 2 != 0) {
8             System.out.println("Weird");
9         } else if (N >= 2 && N <= 5) {
10            System.out.println("Not Weird");
11        } else if (N >= 6 && N <= 20) {
12            System.out.println("Weird");
13        } else {
14            System.out.println("Not Weird");
15        }
16        sc.close();
17    }
18 }
```

Line: 18 Col: 1

Upload Code as File Test against custom input

Run Code Submit Code

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Search

ENG IN

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3.

hackerank.com/challenges/java-stdin-stdout/problem?isFullScreen=true

HackerRank | Prepare | Java | Introduction | Java Stdin and Stdout II

Exit Full Screen View

Problem

In this challenge, you must read an integer, a double, and a String from stdin, then print the values according to the instructions in the Output Format section below. To make the problem a little easier, a portion of the code is provided for you in the editor.

Note: We recommend completing [Java Stdin and Stdout I](#) before attempting this challenge.

Input Format

There are three lines of input:

1. The first line contains an integer.
2. The second line contains a double.
3. The third line contains a String.

Output Format

There are three lines of output:

1. On the first line, print `String`: followed by the unaltered String read from stdin.
2. On the second line, print `Double`: followed by the unaltered double read from stdin.
3. On the third line, print `Int`: followed by the unaltered integer read from stdin.

To make the problem easier, a portion of the code is already provided in the editor.

Note: If you use the `nextLine()` method immediately following the `nextInt()` method, recall that `nextInt()` reads integer tokens; because of this, the last newline character for that line of integer input is still queued in the input buffer and the next `nextLine()` will be reading the remainder of the integer line (which is empty).

Sample Input

```
1
2.5
String
```

Submissions

Leaderboard

Discussions

Change Theme Language Java 15

```
1 import java.util.Scanner;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int i = sc.nextInt();
6         double d = sc.nextDouble();
7         sc.nextLine();
8         String s = sc.nextLine();
9         System.out.println("String: " + s);
10        System.out.println("Double: " + d);
11        System.out.println("Int: " + i);
12    }
13 }
14
15
16
```

Line: 16 Col: 1

Upload Code as File Test against custom input

Run Code Submit Code

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4.

hackerank.com/challenges/java-output-formatting/problem?isFullScreen=true

HackerRank | Prepare | Java | Introduction | Java Output Formatting

Exit Full Screen View

Problem

Java's `System.out.printf` function can be used to print formatted output. The purpose of this exercise is to test your understanding of formatting output using `printf`.

To get you started, a portion of the solution is provided for you in the editor; you must format and print the input to complete the solution.

Input Format

Every line of input will contain a String followed by an integer.

Each String will have a maximum of 10 alphabetic characters, and each integer will be in the inclusive range from 0 to 999.

Output Format

In each line of output there should be two columns:

The first column contains the String and is left justified using exactly 15 characters.

The second column contains the integer, expressed in exactly 3 digits; if the original input has less than three digits, you must pad your output's leading digits with zeroes.

Sample Input

```
java 100
cpp 65
python 50
```

Sample Output

```
java 100
cpp 65
python 50
```

Submissions

Leaderboard

Discussions

Change Theme Language Java 15

```
1 import java.util.Scanner;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.println("=====");
6         for (int i = 0; i < 3; i++) {
7             String s = sc.next();
8             int x = sc.nextInt();
9             System.out.printf("%-15s%03d\n", s, x);
10        }
11        System.out.println("=====");
12        sc.close();
13    }
14 }
15
```

Line: 15 Col: 1

Upload Code as File Test against custom input

Run Code Submit Code

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5.

hackerank.com/challenges/java-loops-1/problem?isFullScreen=true

HackerRank | Prepare | Java | Introduction | Java Loops I | Exit Full Screen View

Problem

Objective
In this challenge, we're going to use loops to help us do some simple math.

Task
Given an integer, N , print its first 10 multiples. Each multiple $N \times i$ (where $1 \leq i \leq 10$) should be printed on a new line in the form: $N \times i = \text{result}$.

Input Format
A single integer, N .

Constraints
 $2 \leq N \leq 20$

Output Format
Print 10 lines of output; each line i (where $1 \leq i \leq 10$) contains the **result** of $N \times i$ in the form:
 $N \times i = \text{result}$.

Sample Input
2

Sample Output
2 x 1 = 2
2 x 2 = 4

```

1 import java.util.Scanner;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int N = sc.nextInt();
6         for (int i = 1; i <= 10; i++) {
7             System.out.println(N + " x " + i + " = " + (N * i));
8         }
9         sc.close();
10    }
11 }
12

```

Line: 12 Col: 1

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6.

hackerank.com/challenges/java-loops-2/problem?isFullScreen=true

HackerRank | Prepare | Java | Introduction | Java Loops II | Exit Full Screen View

Problem

We use the integers a , b , and n to create the following series:
 $(a + 2^0 \cdot b), (a + 2^0 \cdot b + 2^1 \cdot b), \dots, (a + 2^0 \cdot b + 2^1 \cdot b + \dots + 2^{n-1} \cdot b)$

You are given q queries in the form of a , b , and n . For each query, print the series corresponding to the given a , b , and n values as a single line of n space-separated integers.

Input Format
The first line contains an integer, q , denoting the number of queries.
Each line i of the q subsequent lines contains three space-separated integers describing the respective a_i , b_i , and n_i values for that query.

Constraints

- $0 \leq q \leq 500$
- $0 \leq a, b \leq 50$
- $1 \leq n \leq 15$

Output Format
For each query, print the corresponding series on a new line. Each series must be printed in order as a single line of n space-separated integers.

Sample Input
2
0 2 10
5 3 5

```

1 import java.util.Scanner;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int q = sc.nextInt();
6         while (q-- > 0) {
7             int a = sc.nextInt();
8             int b = sc.nextInt();
9             int n = sc.nextInt();
10            int sum = a;
11            for (int i = 0; i < n; i++) {
12                sum += (1 << i) * b;
13                System.out.print(sum + " ");
14            }
15            System.out.println();
16        }
17        sc.close();
18    }
19 }
20

```

Line: 20 Col: 1

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7.

hackerank.com/challenges/java-datatypes/problem?isFullScreen=true

HackerRank | Prepare | Java | Introduction | Java Datatypes | Exit Full Screen View

Problem

Java has 8 primitive data types: char, boolean, byte, short, int, long, float, and double. For this exercise, we'll work with the primitives used to hold integer values (byte, short, int, and long):

- A byte is an 8-bit signed integer.
- A short is a 16-bit signed integer.
- An int is a 32-bit signed integer.
- A long is a 64-bit signed integer.

Given an input integer, you must determine which primitive data types are capable of properly storing that input.

To get you started, a portion of the solution is provided for you in the editor.

Reference: <https://docs.oracle.com/javase/tutorial/java/nutsandbolts/datatypes.html>

Input Format

The first line contains an integer, T , denoting the number of test cases.

Each test case, T , is comprised of a single line with an integer, n , which can be arbitrarily large or small.

Output Format

For each input variable n and appropriate primitive **dataType**, you must determine if the given primitives are capable of storing it. If yes, then print:

```
n can be fitted in:
* dataType
```

Submissions

Leaderboard

Discussions

```
1 import java.util.Scanner;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int t = sc.nextInt();
6         for (int i = 0; i < t; i++) {
7             try {
8                 long x = sc.nextLong();
9                 System.out.println(x + " can be fitted in:");
10                if (x >= Byte.MIN_VALUE && x <= Byte.MAX_VALUE)
11                    System.out.println(" byte");
12                if (x >= Short.MIN_VALUE && x <= Short.MAX_VALUE)
13                    System.out.println(" short");
14                if (x >= Integer.MIN_VALUE && x <= Integer.MAX_VALUE)
15                    System.out.println(" int");
16                System.out.println(" long");
17            } catch (Exception e) {
18                System.out.println(sc.next() + " can't be fitted anywhere.");
19            }
20        }
21    }
22    sc.close();
23 }
24 }
25
```

Line: 25 Col: 1

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8.

hackerank.com/challenges/java-end-of-file/problem?isFullScreen=true

HackerRank | Prepare | Java | Introduction | Java End-of-file | Exit Full Screen View

Problem

"In computing, End Of File (commonly abbreviated EOF) is a condition in a computer operating system where no more data can be read from a data source." — (Wikipedia: End-of-file)

The challenge here is to read n lines of input until you reach EOF, then number and print all n lines of content.

Hint: Java's Scanner.hasNext() method is helpful for this problem.

Input Format

Read some unknown n lines of input from stdin(System.in) until you reach EOF; each line of input contains a non-empty String.

Output Format

For each line, print the line number, followed by a single space, and then the line content received as input.

Sample Input

```
Hello world
I am a file
Read me until end-of-file.
```

Sample Output

```
1 Hello world
```

Submissions

Leaderboard

Discussions

```
1 import java.util.Scanner;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int lineNumber = 1;
6         while (sc.hasNextLine()) {
7             String line = sc.nextLine();
8             System.out.println(lineNumber + " " + line);
9             lineNumber++;
10        }
11    }
12    sc.close();
13 }
14
```

Line: 14 Col: 1

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9.

Static initialization blocks are executed when the class is loaded, and you can initialize static variables in those blocks.

It's time to test your knowledge of Static initialization blocks. You can read about it [here](#).

You are given a class `Solution` with a main method. Complete the given code so that it outputs the area of a parallelogram with breadth B and height H . You should read the variables from the standard input.

If $B \leq 0$ or $H \leq 0$, the output should be "java.lang.Exception: Breadth and height must be positive" without quotes.

Input Format

There are two lines of input. The first line contains B : the breadth of the parallelogram. The next line contains H : the height of the parallelogram.

Constraints

- $-100 \leq B \leq 100$
- $-100 \leq H \leq 100$

Output Format

If both values are greater than zero, then the main method must output the area of the parallelogram. Otherwise, print "java.lang.Exception: Breadth and height must be positive" without quotes.

Sample input 1

```

1  import java.util.Scanner;
2  public class Solution {
3      static int B;
4      static int H;
5      static boolean flag = true;
6      static {
7          Scanner sc = new Scanner(System.in);
8          B = sc.nextInt();
9          H = sc.nextInt();
10         if (B <= 0 || H <= 0) {
11             flag = false;
12             System.out.println("java.lang.Exception: Breadth and height must be positive");
13         }
14         sc.close();
15     }
16     public static void main(String[] args) {
17         if (flag) {
18             int area = B * H;
19             System.out.println(area);
20         }
21     }
22 }

```

Line: 16 Col: 45

Run Code Submit Code

11

The `Calendar` class is an abstract class that provides methods for converting between a specific instant in time and a set of calendar fields such as YEAR, MONTH, DAY_OF_MONTH, HOUR, and so on, and for manipulating the calendar fields, such as getting the date of the next week.

You are given a date. You just need to write the method, `getDay`, which returns the day on that date. To simplify your task, we have provided a portion of the code in the editor.

Example

month = 8
day = 14
year = 2017

The method should return `MONDAY` as the day on that date.

AUGUST 2017

SUN	MON	TUE	WED	THU	FRI	SAT
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19

```

1  import java.util.*;
2  import java.time.*;
3  public class Solution {
4      public static String findDay(int month, int day, int year) {
5          LocalDate date = LocalDate.of(year, month, day);
6          return date.getDayOfWeek().toString();
7      }
8      public static void main(String[] args) {
9          Scanner sc = new Scanner(System.in);
10         int month = sc.nextInt();
11         int day = sc.nextInt();
12         int year = sc.nextInt();
13         System.out.println(findDay(month, day, year));
14         sc.close();
15     }
16 }

```

Line: 17 Col: 1

Run Code Submit Code

10.

hackerank.com/challenges/java-int-to-string/problem?isFullScreen=true

HackerRank | Prepare | Java | Introduction | Java Int to String | Exit Full Screen View

Problem

You are given an integer n , you have to convert it into a string.

Please complete the partially completed code in the editor. If your code successfully converts n into a string s the code will print "Good job". Otherwise it will print "Wrong answer".

n can range between -100 to 100 inclusive.

Sample Input 0

```
100
```

Sample Output 0

```
Good job
```

Submissions

Leaderboard

Discussions

```

1  import java.util.*;
2  public class Solution {
3      public static void main(String[] args) {
4          Scanner sc = new Scanner(System.in);
5          int n = sc.nextInt();
6          sc.close();
7          String s = Integer.toString(n);
8          if (n == Integer.parseInt(s)) {
9              System.out.println("Good job");
10         } else {
11             System.out.println("Wrong answer");
12         }
13     }
14 }
15

```

Line: 15 Col: 1

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12

hackerank.com/challenges/java-strings-introduction/problem?isFullScreen=true

HackerRank | Prepare | Java | Strings | Java Strings Introduction | Exit Full Screen View

Problem

"A string is traditionally a sequence of characters, either as a literal constant or as some kind of variable." — Wikipedia: String (computer science)

This exercise is to test your understanding of Java Strings. A sample String declaration:

```
String myString = "Hello World!"
```

The elements of a String are called characters. The number of characters in a String is called the length, and it can be retrieved with the String.length() method.

Given two strings of lowercase English letters, A and B , perform the following operations:

- Sum the lengths of A and B .
- Determine if A is lexicographically larger than B (i.e.: does B come before A in the dictionary?).
- Capitalize the first letter in A and B and print them on a single line, separated by a space.

Input Format

The first line contains a string A . The second line contains another string B . The strings are comprised of only lowercase English letters.

Output Format

There are three lines of output:

For the first line, sum the lengths of A and B .

For the second line, write Yes if A is lexicographically greater than B otherwise print No.

Submissions

Leaderboard

Discussions

```

1  import java.util.Scanner;
2  public class Solution {
3      public static void main(String[] args) {
4          Scanner sc = new Scanner(System.in);
5          String A = sc.next();
6          String B = sc.next();
7          System.out.println(A.length() + B.length());
8          if (A.compareTo(B) > 0)
9              System.out.println("Yes");
10         else
11             System.out.println("No");
12         A = A.substring(0, 1).toUpperCase() + A.substring(1);
13         B = B.substring(0, 1).toUpperCase() + B.substring(1);
14         System.out.println(A + " " + B);
15         sc.close();
16     }
17 }
18

```

Line: 18 Col: 1

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13.

hackerank.com/challenges/java-string-tokens/problem?isFullScreen=true

HackerRank | Prepare | Java | Strings | Java String Tokens | Exit Full Screen View

Problem

Given a string, s , matching the regular expression $[A-Za-z \, ? \, _ \, @]^+$, split the string into tokens. We define a token to be one or more consecutive English alphabetic letters. Then, print the number of tokens, followed by each token on a new line.

Note: You may find the `String.split` method helpful in completing this challenge.

Input Format

A single string, s .

Constraints

- $1 \leq \text{length of } s \leq 4 \cdot 10^5$
- s is composed of any of the following: English alphabetic letters, blank spaces, exclamation points (!), commas (,), question marks (?), periods (.), underscores (_), apostrophes ('), and at symbols (@).

Output Format

On the first line, print an integer, n , denoting the number of tokens in string s (they do not need to be unique). Next, print each of the n tokens on a new line in the same order as they appear in input string s .

Sample Input

```
He is a very very good boy, isn't he?
```

Sample Output

```
19
He
is
a
very
very
good
boy,
isn't
he?
```

```
1 import java.util.Scanner;
2 public class Solution {
3     public static void main(String[] args) {
4         Scanner scan = new Scanner(System.in);
5         String s = scan.nextLine();
6         scan.close();
7         s = s.trim();
8         if (s.length() == 0) {
9             System.out.println(0);
10            return;
11        }
12        String[] tokens = s.split("[ !,?._@]+");
13        System.out.println(tokens.length);
14        for (String token : tokens) {
15            System.out.println(token);
16        }
17    }
18 }
19
```

Line: 19 Col: 1

Upload Codes as File Test against custom input Run Code Submit Code

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14.

hackerank.com/challenges/pattern-syntax-checker/problem?isFullScreen=true

HackerRank | Prepare | Java | Strings | Pattern Syntax Checker | Exit Full Screen View

Problem

Using **Regex**, we can easily match or search for patterns in a text. Before searching for a pattern, we have to specify one using some well-defined syntax.

In this problem, you are given a pattern. You have to check whether the syntax of the given pattern is valid.

Note: In this problem, a regex is only valid if you can compile it using the `Pattern.compile` method.

Input Format

The first line of input contains an integer N , denoting the number of test cases. The next N lines contain a string of any printable characters representing the pattern of a regex.

Output Format

For each test case, print `Valid` if the syntax of the given pattern is correct. Otherwise, print `Invalid`. Do not print the quotes.

Sample Input

```
3
{[A-Z]}(.+ )
[AZ[a-z]{a-z}
batcatpat(nat
```

Sample Output

```
Valid
Invalid
Invalid
```

```
1 import java.util.Scanner;
2 import java.util.regex.Pattern;
3 import java.util.regex.PatternSyntaxException;
4 public class Solution {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         int testCases = Integer.parseInt(sc.nextLine());
8         while (testCases-- > 0) {
9             String pattern = sc.nextLine();
10            try {
11                Pattern.compile(pattern);
12                System.out.println("Valid");
13            } catch (PatternSyntaxException e) {
14                System.out.println("Invalid");
15            }
16        }
17        sc.close();
18    }
19 }
20
```

Line: 20 Col: 2

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15.

hackerank.com/challenges/java-primality-test/problem?isFullScreen=true

HackerRank | Prepare | Java | **BigNumber** | Java Primality Test

Problem

A prime number is a natural number greater than 1 whose only positive divisors are 1 and itself. For example, the first six prime numbers are 2, 3, 5, 7, 11, and 13.

Given a large integer, n , use the Java BigInteger class' `isProbablePrime` method to determine and print whether it's prime or not prime.

Input Format

A single line containing an integer, n (the number to be checked).

Constraints

- n contains at most 100 digits.

Output Format

If n is a prime number, print `prime`; otherwise, print `not prime`.

Sample Input

```
13
```

Sample Output

```
prime
```

Explanation

The only positive divisors of 13 are 1 and 13, so we print `prime`.

Change Theme Language Java 15

```
1 import java.math.BigInteger;
2 import java.util.Scanner;
3 public class Solution {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         BigInteger n = sc.nextBigInteger();
7         if (n.isProbablePrime(100)) {
8             System.out.println("prime");
9         } else {
10            System.out.println("not prime");
11        }
12        sc.close();
13    }
14 }
15
```

Line: 15 Col: 1

Upload Code as File Test against custom input

Run Code Submit Code

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16.

hackerank.com/challenges/java-generics/problem?isFullScreen=true

HackerRank | Prepare | Java | **Data Structures** | Java Generics

Problem

Generic methods are a very efficient way to handle multiple datatypes using a single method. This problem will test your knowledge on Java Generic methods.

Let's say you have an integer array and a string array. You have to write a **single** method `printArray` that can print all the elements of both arrays. The method should be able to accept both integer arrays or string arrays.

You are given code in the editor. Complete the code so that it prints the following lines:

```
1
2
3
Hello
World
```

Do not use method overloading because your answer will not be accepted.

Change Theme Language Java 15

```
1 public class Solution {
2     static class Printer {
3         public <T> void printArray(T[] array) {
4             for (T element : array) {
5                 System.out.println(element);
6             }
7         }
8     }
9     public static void main(String[] args) {
10        Printer printer = new Printer();
11        Integer[] intArray = {1, 2, 3};
12        String[] stringArray = {"Hello", "World"};
13        printer.printArray(intArray);
14        printer.printArray(stringArray);
15    }
16 }
17
```

Line: 17 Col: 1

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17.

hackerank.com/challenges/java-bitset/problem?isFullScreen=true

HackerRank | Prepare | Java | Data Structures | Java BitSet

Java's `BitSet` class implements a vector of bit values (i.e.: `false` (0) or `true` (1)) that grows as needed, allowing us to easily manipulate bits while optimizing space (when compared to other collections). Any element having a bit value of 1 is called a set bit.

Given 2 `BitSets`, B_1 and B_2 , of size N where all bits in both `BitSets` are initialized to 0, perform a series of M operations. After each operation, print the number of set bits in the respective `BitSets` as two space-separated integers on a new line.

Input Format

The first line contains 2 space-separated integers, N (the length of both `BitSets` B_1 and B_2) and M (the number of operations to perform), respectively.

The M subsequent lines each contain an operation in one of the following forms:

- `AND <set> <set>`
- `OR <set> <set>`
- `XOR <set> <set>`
- `FLIP <set> <index>`
- `SET <set> <index>`

In the list above, `<set>` is the integer 1 or 2, where 1 denotes B_1 and 2 denotes B_2 . `<index>` is an integer denoting a bit's index in the `BitSet` corresponding to `<set>`.

For the binary operations `AND`, `OR`, and `XOR`, operands are read from left to right and the `BitSet` resulting from the operation replaces the contents of the first operand. For example:

```
AND 2 1
```

1 import java.util.*;
2 public class Solution {
3 public static void main(String[] args) {
4 Scanner sc = new Scanner(System.in);
5 int n = sc.nextInt();
6 int m = sc.nextInt();
7 BitSet b1 = new BitSet(n);
8 BitSet b2 = new BitSet(n);
9 for (int i = 0; i < m; i++) {
10 String operation = sc.next();
11 int set = sc.nextInt();
12 int value = sc.nextInt();
13 switch (operation) {
14 case "AND":
15 if (set == 1)
16 b1.and(b2);
17 else
18 b2.and(b1);
19 break;
20 case "OR":
21 if (set == 1)
22 b1.or(b2);
23 else
24 b2.or(b1);
25 break;

Line: 50 Col: 1

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18.

hackerank.com/challenges/java-iterator/problem?isFullScreen=true

HackerRank | Prepare | Java | Object Oriented Programming | Java Iterator

Java `Iterator` class can help you to iterate through every element in a collection. Here is a simple example:

```
import java.util.*;  
public class Example {  
  
    public static void main(String []args) {  
        ArrayList mylist = new ArrayList();  
        mylist.add("Hello");  
        mylist.add("Java");  
        mylist.add("4");  
        Iterator it = mylist.iterator();  
        while(it.hasNext()){  
            Object element = it.next();  
            System.out.println(String.valueOf(element));  
        }  
    }  
}
```

In this problem you need to complete a method `func`. The method takes an `ArrayList` as input. In that `ArrayList` there is one or more integer numbers, then there is a special string `###`, after that there are one or more other strings. A sample `ArrayList` may look like this:

```
element[0] => 42  
element[1] => 10  
element[2] => "###"  
element[3] => "Hello"  
element[4] => "Java"
```

1 import java.util.*;
2 public class Solution {
3 static Iterator func(ArrayList mylist) {
4 Iterator it = mylist.iterator();
5 while (it.hasNext()) {
6 Object element = it.next();
7 if (element.equals("###")) {
8 break;
9 }
10 }
11 return it;
12 }
13 public static void main(String[] args) {
14 ArrayList mylist = new ArrayList();
15 Scanner sc = new Scanner(System.in);
16 int n = sc.nextInt();
17 int m = sc.nextInt();
18 for (int i = 0; i < n; i++) {
19 mylist.add(sc.nextInt());
20 }
21 mylist.add("###");
22 for (int i = 0; i < m; i++) {
23 mylist.add(sc.next());
24 }
25 Iterator it = func(mylist);

Line: 32 Col: 1

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19.

hackerank.com/challenges/java-exception-handling/problem?isFullScreen=true

HackerRank | Prepare Java Exception Handling Java Exception Handling Exit Full Screen View

Problem

You are required to compute the power of a number by implementing a calculator. Create a class `MyCalculator` which consists of a single method `long power(int, int)`. This method takes two integers, `n` and `p`, as parameters and finds n^p . If either `n` or `p` is negative, then the method must throw an exception which says "n or p should not be negative". Also, if both `n` and `p` are zero, then the method must throw an exception which says "n and p should not be zero."

For example, -4 and -5 would result in

java.lang.Exception: n or p should not be negative.

Complete the function `power` in class `MyCalculator` and return the appropriate result after the power operation or an appropriate exception as detailed above.

Input Format

Each line of the input contains two integers, `n` and `p`. The locked stub code in the editor reads the input and sends the values to the method as parameters.

Constraints

- $-10 \leq n \leq 10$
- $-10 \leq p \leq 10$

Output Format

Each line of the output contains the result n^p , if both `n` and `p` are positive. If either `n` or `p` is negative, the output contains "n and p should be non-negative". If both `n` and `p` are zero, the output contains "n and p should not be zero.". This is printed by the locked stub code in the editor.

Change Theme Language Java 15

```
1 import java.util.Scanner;
2 class MyCalculator {
3     long power(int n, int p) throws Exception {
4         if (n < 0 || p < 0) {
5             throw new Exception("n or p should not be negative.");
6         }
7         if (n == 0 && p == 0) {
8             throw new Exception("n and p should not be zero.");
9         }
10        return (long) Math.pow(n, p);
11    }
12 }
13 public class Solution {
14     public static void main(String[] args) {
15         Scanner sc = new Scanner(System.in);
16         MyCalculator myCalculator = new MyCalculator();
17         while (sc.hasNextInt()) {
18             int n = sc.nextInt();
19             int p = sc.nextInt();
20             try {
21                 System.out.println(myCalculator.power(n, p));
22             } catch (Exception e) {
23                 System.out.println(e);
24             }
25         }
26     }
27 }
```

Line: 29 Col: 1

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20.

hackerank.com/challenges/java-singleton/problem?isFullScreen=true

HackerRank | Prepare Java Advanced Java Singleton Pattern Exit Full Screen View

Problem

"The singleton pattern is a design pattern that restricts the instantiation of a class to one object. This is useful when exactly one object is needed to coordinate actions across the system."

- Wikipedia: Singleton Pattern

Complete the Singleton class in your editor which contains the following components:

- A private Singleton non parameterized constructor.
- A public String instance variable named `str`.
- Write a static method named `getSingletonInstance` that returns the single instance of the Singleton class.

Once submitted, our hidden Solution class will check your code by taking a String as input and then using your Singleton class to print a line.

Input Format

You will not be handling any input in this challenge.

Output Format

You will not be producing any output in this challenge.

Sample Input

hello world

Sample Output

Change Theme Language Java 15

```
1 import java.util.Scanner;
2 class Singleton {
3     public String str;
4     private static Singleton singleInstance = new Singleton();
5     private Singleton() {
6     }
7     public static Singleton getSingletonInstance() {
8         return singleInstance;
9     }
10 }
11 public class Solution {
12     public static void main(String[] args) {
13         Scanner sc = new Scanner(System.in);
14         String s = sc.nextLine();
15         Singleton singleton = Singleton.getSingletonInstance();
16         singleton.str = s;
17         System.out.println("Hello I am a singleton! Let me say " + singleton.str
18             " to you");
19         sc.close();
20     }
21 }
```

Line: 21 Col: 1

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